# Kernix 0.8.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

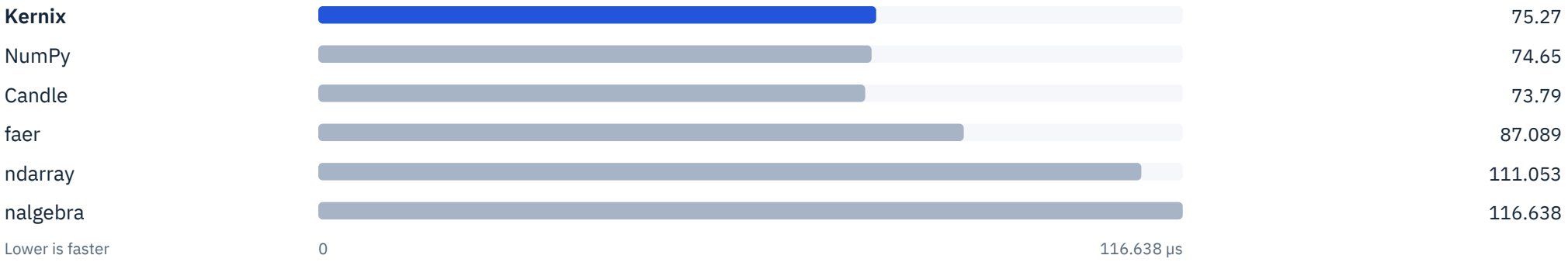
**21** workloads

**88** observations

Source precision retained

## Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

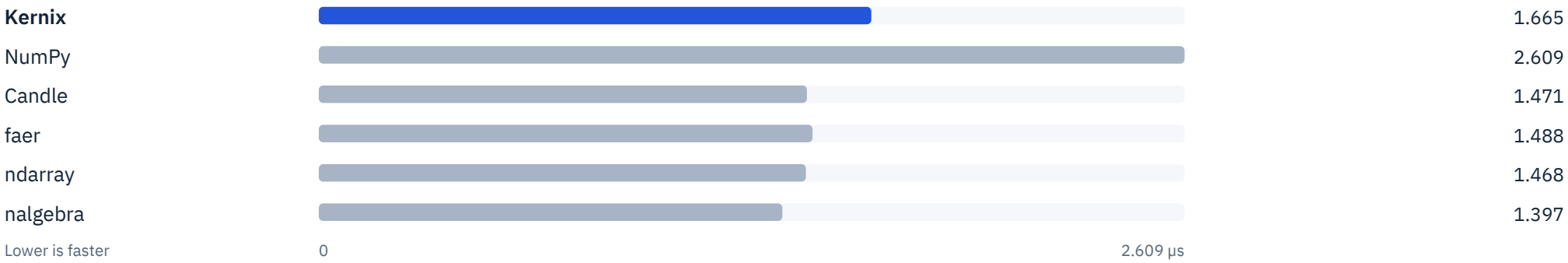
# Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

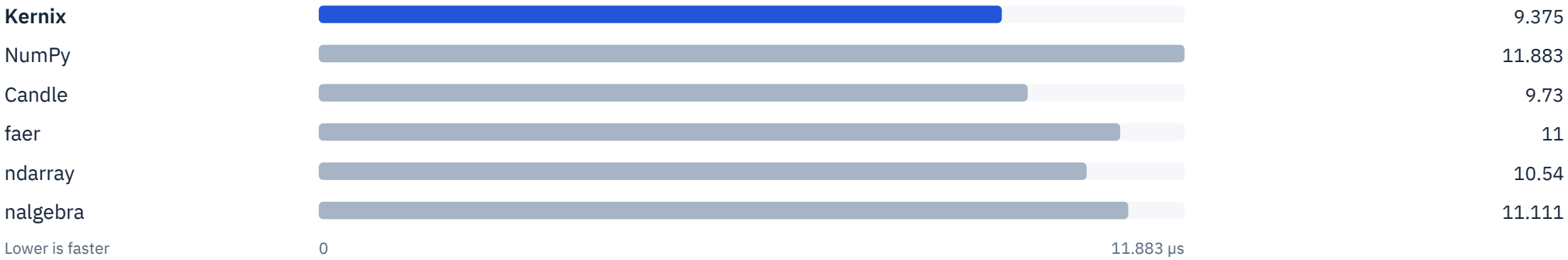
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



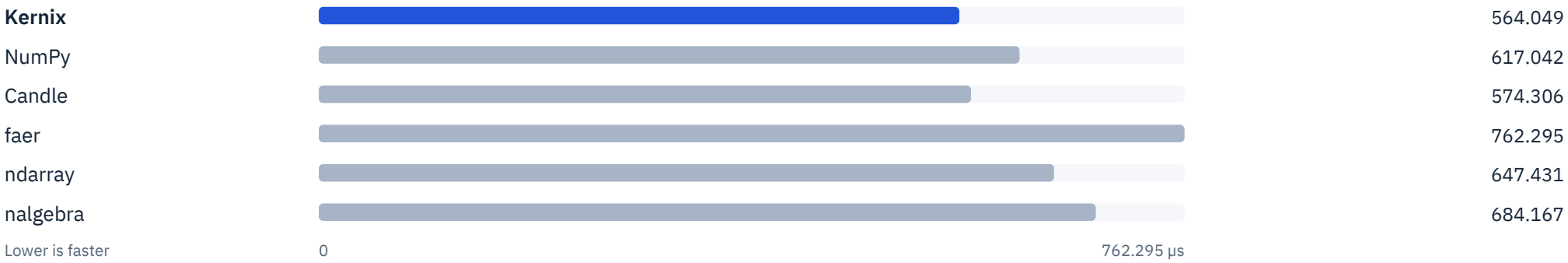
# Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

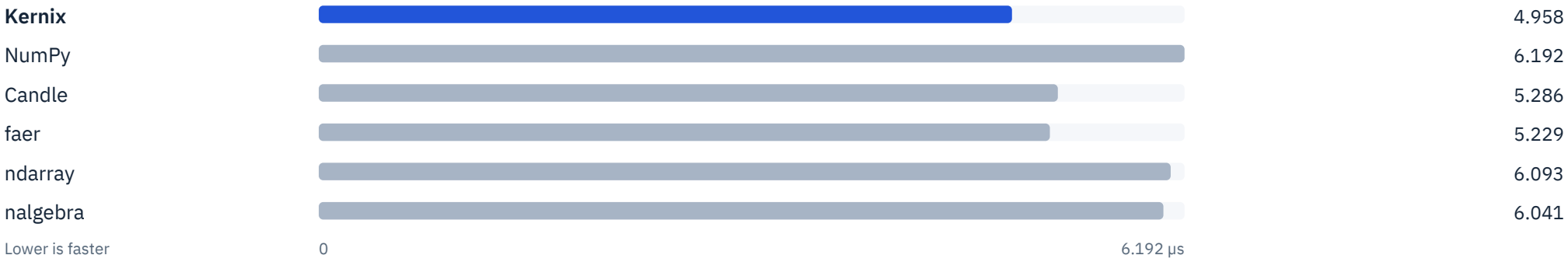
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



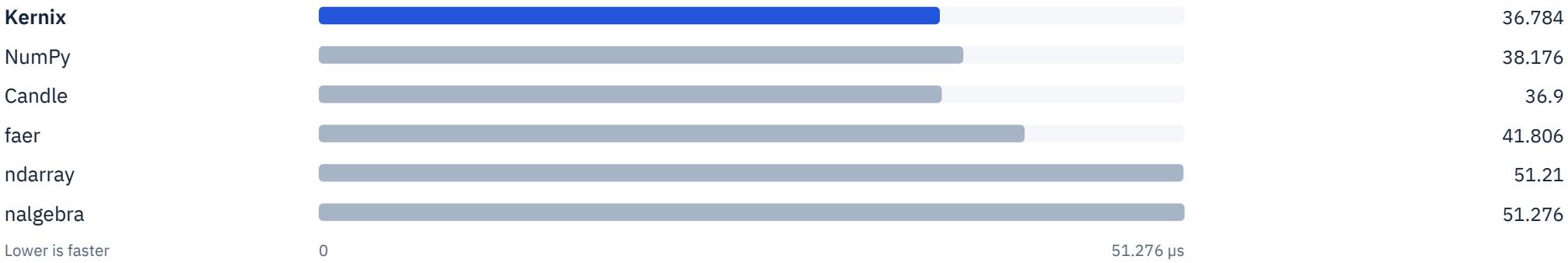
# Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

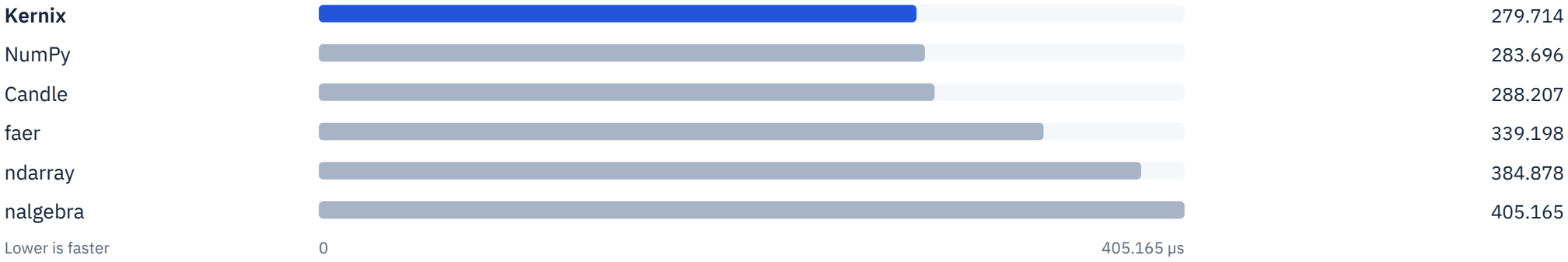
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

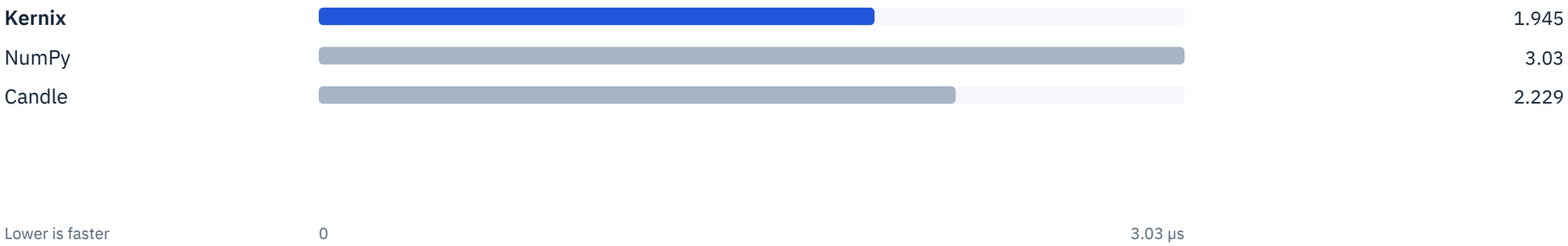


# Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

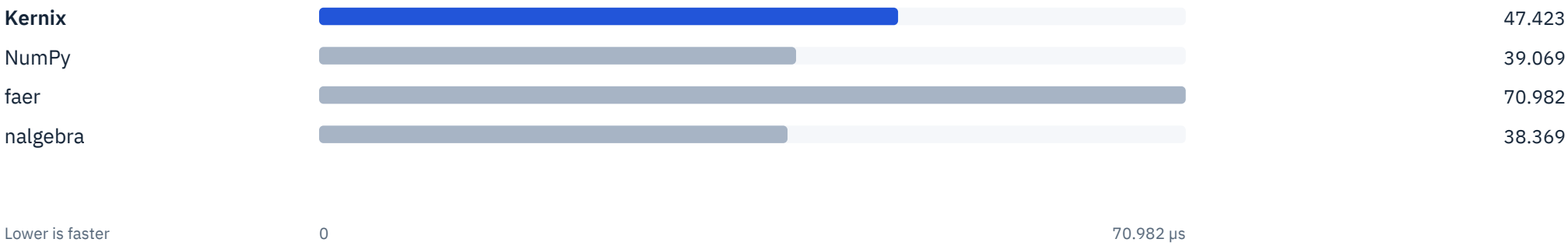


# Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Thin QR

96 × 32 · f64 · 1 worker · allocated output




## Thin QR

256 × 64 · f64 · 1 worker · allocated output



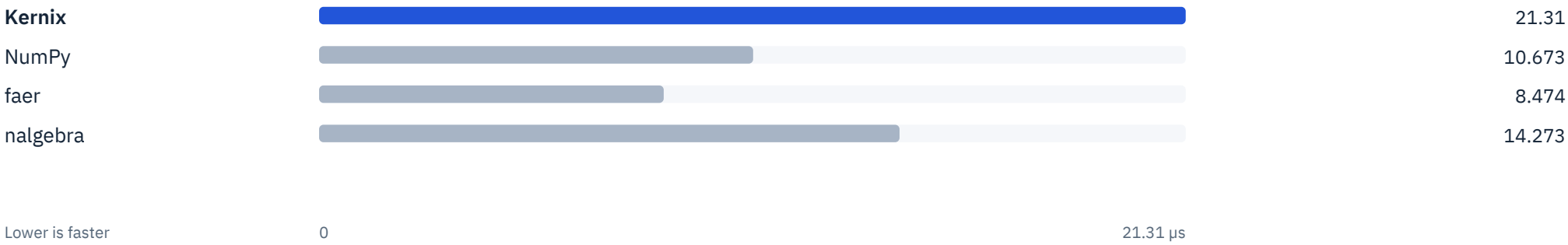
# Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

Per-call median

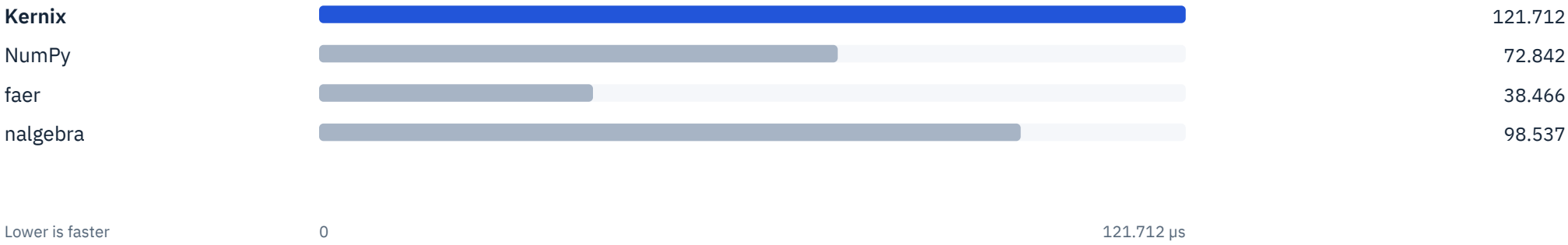
64 × 64 · f64 · 1 worker · allocated output




## Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

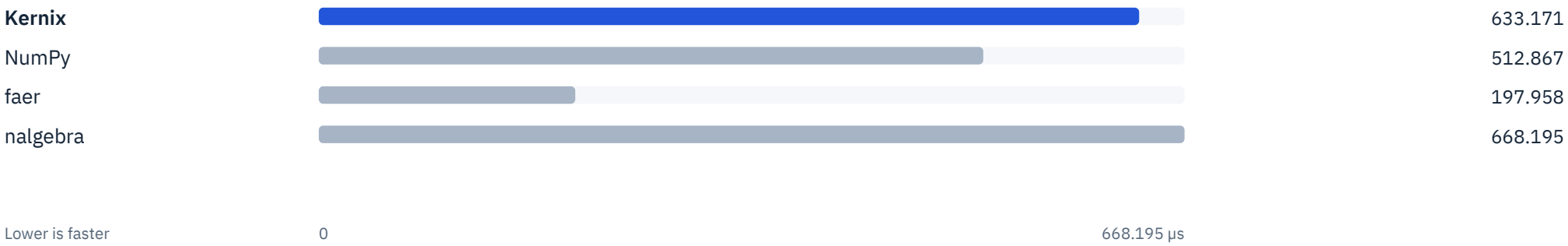


# Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



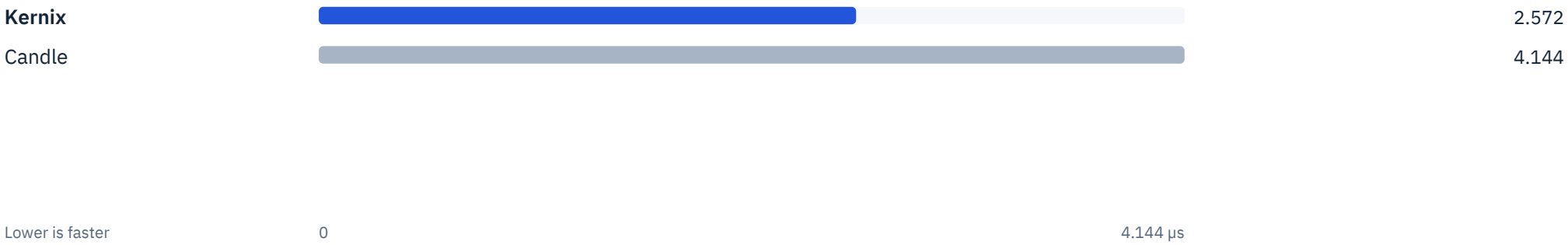
# Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Record + backward

Per-call median

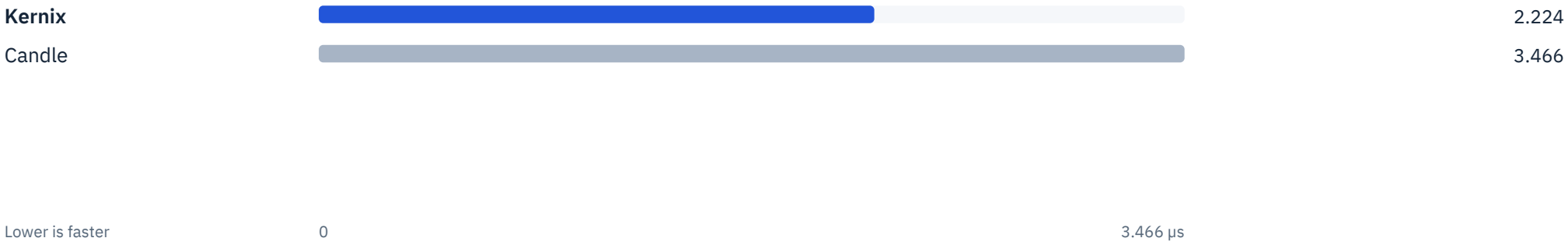
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product




## Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model recording + forward

Per-call median

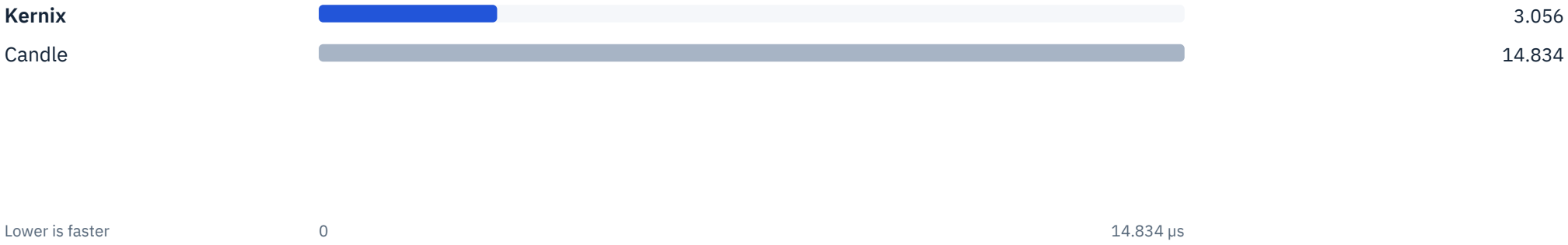
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



# Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

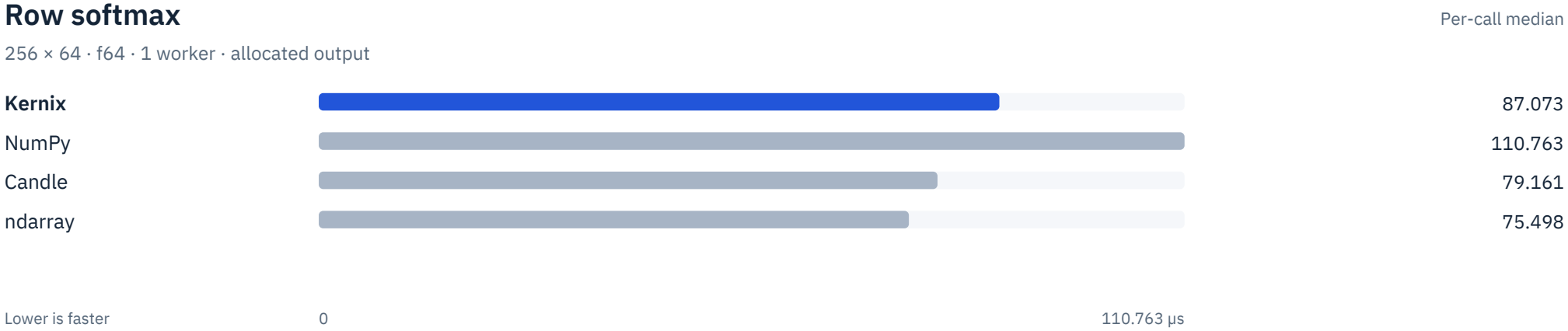


# Kernix 0.8.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output



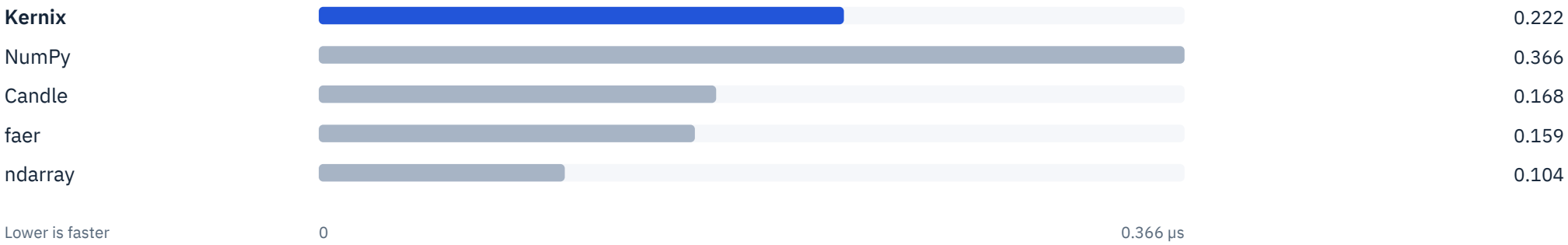
# Kernix 0.8.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median




## Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

64 × 64 · f64

**Observed time reduction**

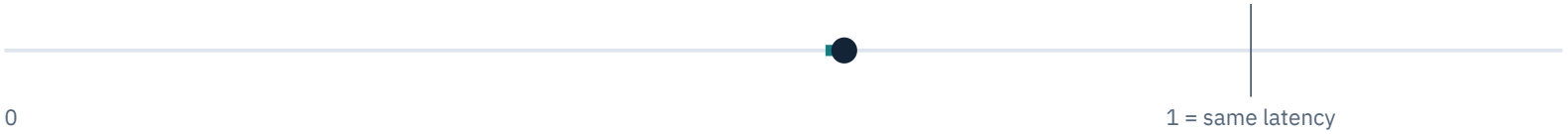



| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.8677118475888073** | 0.826547772693496 – 0.9410034528326016 |

## Cholesky factorization

128 × 128 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.6737598467695953** | 0.6587561862739252 – 0.6826653795859267 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

256 × 256 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.49316287663357755** | 0.4813219013084302 – 0.5031802131103422 |

## Thin QR

96 × 32 · f64

**Observed time increase**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0433176701999782** | 1.036087431693989 – 1.0842891742448475 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Thin QR

256 × 64 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9581048304614215** | 0.9368562130504658 – 0.9876960858970454 |

## Matrix multiplication

128 × 128 · 128 × 128 · f64

**Observed time increase**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0148811150587593** | 1.0060131963811985 – 1.0317538528920642 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| | | | |
|---|---|---|---|
| CURRENT / BASELINE | **0.9939454550368949** | 95% INTERVAL | 0.9249689243870831 – 1.0027472527472527 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1665 | 1.665 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 1468 | 1.468 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 1488 | 1.488 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 1471 | 1.471 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 2609 | 2.609 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 1397 | 1.397 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 11111 | 11.111 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 11883 | 11.883 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 9375 | 9.375 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 10540 | 10.54 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 11000 | 11 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 9730 | 9.73 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 116638 | 116.638 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 87089 | 87.089 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 111053 | 111.053 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 75270 | 75.27 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 74650 | 74.65 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 73790 | 73.79 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 684167 | 684.167 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 762295 | 762.295 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 617042 | 617.042 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 564049 | 564.049 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 574306 | 574.306 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 647431 | 647.431 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f32 | 6041 | 6.041 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f32 | 5286 | 5.286 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f32 | 4958 | 4.958 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f32 | 6192 | 6.192 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f32 | 5229 | 5.229 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f32 | 6093 | 6.093 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f32 | 51210 | 51.21 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f32 | 38176 | 38.176 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f32 | 51276 | 51.276 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f32 | 36784 | 36.784 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f32 | 36900 | 36.9 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f32 | 41806 | 41.806 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f32 | 339198 | 339.198 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f32 | 288207 | 288.207 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f32 | 405165 | 405.165 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f32 | 279714 | 279.714 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f32 | 283696 | 283.696 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f32 | 384878 | 384.878 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 2229 | 2.229 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 3030 | 3.03 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1945 | 1.945 |
| **Thin QR / Kernix**<br>96 × 32 · f64 | 47423 | 47.423 |
| **Thin QR / faer**<br>96 × 32 · f64 | 70982 | 70.982 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 38369 | 38.369 |
| **Thin QR / NumPy**<br>96 × 32 · f64 | 39069 | 39.069 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 484243 | 484.243 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / faer**<br>256 × 64 · f64 | 416558 | 416.558 |
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 372930 | 372.93 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 348582 | 348.582 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 14273 | 14.273 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 21310 | 21.31 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 8474 | 8.474 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 10673 | 10.673 |
| **Cholesky factorization / NumPy**<br>128 × 128 · f64 | 72842 | 72.842 |
| **Cholesky factorization / nalgebra**<br>128 × 128 · f64 | 98537 | 98.537 |
| **Cholesky factorization / faer**<br>128 × 128 · f64 | 38466 | 38.466 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / Kernix**<br>128 × 128 · f64 | 121712 | 121.712 |
| **Cholesky factorization / faer**<br>256 × 256 · f64 | 197958 | 197.958 |
| **Cholesky factorization / nalgebra**<br>256 × 256 · f64 | 668195 | 668.195 |
| **Cholesky factorization / Kernix**<br>256 × 256 · f64 | 633171 | 633.171 |
| **Cholesky factorization / NumPy**<br>256 × 256 · f64 | 512867 | 512.867 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 2572 | 2.572 |
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 4144 | 4.144 |
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 3466 | 3.466 |
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 2224 | 2.224 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 2720 | 2.72 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 5085 | 5.085 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 14834 | 14.834 |
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 3056 | 3.056 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 529 | 0.529 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 110763 | 110.763 |
| **Row softmax / Candle**<br>256 × 64 · f64 | 79161 | 79.161 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 87073 | 87.073 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 75498 | 75.498 |
| **Create zeros / Kernix**<br>64 × 64 · f32 | 222 | 0.222 |
| **Create zeros / faer**<br>64 × 64 · f32 | 159 | 0.159 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create zeros / ndarray**<br>64 × 64 · f32 | 104 | 0.104 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 366 | 0.366 |
| **Create zeros / Candle**<br>64 × 64 · f32 | 168 | 0.168 |
| **Create ones / faer**<br>64 × 64 · f32 | 203 | 0.203 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 204 | 0.204 |
| **Create ones / Candle**<br>64 × 64 · f32 | 215 | 0.215 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 1349 | 1.349 |
| **Create ones / ndarray**<br>64 × 64 · f32 | 141 | 0.141 |